# PARK, CHOE, VEITCH: THE LINEAR REPRESENTATION HYPOTHESIS

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## 1 Introduction

### The Linear Representation Hypothesis

High-level concepts are represented *linearly*, as directions in some representation space of a model.[1]

[1] Kiho Park, Yo Joong Choe, Victor Veitch, "The Linear Representation Hypothesis and the Geometry of Large Language Models", ICML 2024. https://arxiv.org/abs/2311.03658

### Three meanings of "linear representation"

- **Subspace.** Each concept is represented as a (1-dimensional) subspace.
- **Measurement.** The probability of a concept value can be measured with a linear probe.
- **Intervention.** The value a concept takes on can be changed, without changing other concepts, by adding a suitable steering vector.

### 1.1 Background: How Does an LLM Work?

- Start with the input context $x$: a sequence of $n$ tokens.
- Turn each token into a vector (its token embedding), sometimes together with information about its position. We end up with $n$ vectors in $\mathbf{R}^d$ for some $d$.[2]
- Pass these $n$ vectors through a huge neural network (a stack of Transformer layers) to obtain $n$ new vectors. The $i$-th vector, which can only see tokens $1, \dots, i$, now contains the information used to predict token $i+1$.
- We already know the first $n$ tokens, so to predict token $n+1$ we only need the $n$-th vector. This is $\lambda(x)$ in the paper.
- For each token $y$ in the vocabulary, the model learns an unembedding vector $\gamma(y)$. The score $\lambda(x)^\top \gamma(y)$ is the logit (unnormalized log-probability) of $y$ being token $n+1$.
- Applying the softmax normalization, we get

$$P(y \mid x) = \frac{\exp(\lambda(x)^\top \gamma(y))}{\sum_{y'} \exp(\lambda(x)^\top \gamma(y'))} \propto \exp(\lambda(x)^\top \gamma(y)).$$

- $\lambda(x) \in \Lambda \simeq \mathbf{R}^d$: *embedding vectors*, i.e., representations of contexts.
- $\gamma(y) \in \Gamma \simeq \mathbf{R}^d$: *unembedding vectors*, i.e., representations of output words.

[2] $d = 4096$ for LLaMA-2-7B. Its vocabulary has 32,000 tokens.

## 2 The Linear Representation Hypothesis

### 2.1 Concepts

INTUITION. A concept is any factor of variation that can be changed in isolation. For example, `French⇒English`: the language of the output can be changed without changing its meaning.

- A *concept variable* $W$ is a latent variable that is caused by the context $X$ and acts as a cause of the output $Y$.[3]
- We only consider *binary* concepts with an ordering. E.g., for `male⇒female`, $W = 0$ is male and $W = 1$ is female.
- Each concept defines a set of *counterfactual outputs* $\{Y(W = w)\}$ that differ only in the value of $W$. E.g., for `male⇒female`, $(Y(0), Y(1))$ is a random element of

$$\{(\text{“man”}, \text{“woman”}),\ (\text{“king”}, \text{“queen”}),\ \dots\}.$$

[3] “we assume the value of concepts can be read off deterministically from the sampled output (e.g., the output ‘king’ implies $W = 0$).” (p. 3)

**Definition** (Causal Separability)**.** Two concepts $W$ and $Z$ are *causally separable* if $Y(W = w, Z = z)$ is well-defined for each $w, z$. That is, they can be varied freely and in isolation. We write $Y(w, z)$ for $Y(W = w, Z = z)$.[4]

A concept can be either represented in the embedding space $\Lambda$ or in the unembedding space $\Gamma$.

[4] For instance, $W$ = `English⇒French` (rows) and $Z$ = `male⇒female` (columns) are causally separable:

| | male | female |
|---|---|---|
| English | “king” | “queen” |
| French | “roi” | “reine” |

### 2.2 Unembedding Representations and Measurement

Define the cone of a vector $v$ as $\text{Cone}(v) = \{\alpha v : \alpha > 0\}$.

**Definition 2.1** (Unembedding Representation)**.** We say that $\bar{\gamma}_W$ is an *unembedding representation* of a concept $W$ if

$$\gamma(Y(1)) - \gamma(Y(0)) \in \text{Cone}(\bar{\gamma}_W) \quad \text{almost surely.}$$

- The subspace notion of linear representation: e.g., $\gamma(\text{“queen”}) - \gamma(\text{“king”})$ is parallel to $\gamma(\text{“woman”}) - \gamma(\text{“man”})$.
- $\bar{\gamma}_W$ is unique up to positive scaling.

**Theorem 2.2** (Measurement Representation)**.** *Let $W$ be a concept, and let $\bar{\gamma}_W$ be the unembedding representation of $W$. Then, given any context embedding $\lambda \in \Lambda$,*

$$\text{logit}\, P\big(Y = Y(1) \mid Y \in \{Y(0), Y(1)\}, \lambda\big) = \alpha \lambda^\top \bar{\gamma}_W,$$

*where $\alpha > 0$ is a function of $\{Y(0), Y(1)\}$.*

*Proof.*

$$\begin{aligned}
&\text{logit}\, P\big(Y = Y(1) \mid Y \in \{Y(0), Y(1)\}, \lambda\big) \\
&\quad = \log \frac{P(Y = Y(1) \mid \lambda)}{P(Y = Y(0) \mid \lambda)} = \log \frac{\exp(\lambda^\top \gamma(Y(1)))}{\exp(\lambda^\top \gamma(Y(0)))} \\
&\quad = \lambda^\top \big(\gamma(Y(1)) - \gamma(Y(0))\big) = \alpha \lambda^\top \bar{\gamma}_W,
\end{aligned}$$

where $\alpha$ is a random variable that depends on $\{Y(0), Y(1)\}$. □

Comparison with linear probes[5]

- Same: in both cases, we get a predictor that is linear on the logit scale.
- Different: the unembedding representation does not incorporate any information about correlated but off-target concepts.[6]
- Also? The linear probe can be used in intermediate layers, but umembedding representation is limited to the final layer.

[5] A linear probe is a linear classifier trained on a model's internal activations $h$, with the model itself frozen. In general, it takes the form $p = \phi(w^\top h + b)$, where $\phi$ is a non-linear function and $w, b$ are the learned parameters. For a binary concept, for instance, $\phi$ can simply be the sigmoid $\sigma(t) = 1/(1+e^{-t})$.

[6] "For example, if French text were disproportionately about men, a probe could learn this information (and include it in the representation), but the unembedding representation would not. In this sense, the unembedding representation might be viewed as an ideal probing representation." (p. 3)

### *2.3 Embedding Representations and Intervention*

Each $\lambda(x)$ defines a distribution over concepts. Consider pairs such as

$$\lambda_0 = \lambda(\text{“He is the monarch of England, ”}),$$
$$\lambda_1 = \lambda(\text{“She is the monarch of England, ”}),$$

which induce different distributions on the target concept but the same distribution on all off-target concepts. A concept is embedding-represented if the differences between all such pairs point in a common direction.

**Definition 2.3** (Embedding Representation)**.** We say that $\bar{\lambda}_W$ is an *embedding representation* of a concept $W$ if we have $\lambda_1 - \lambda_0 \in \mathrm{Cone}(\bar{\lambda}_W)$ for any context embeddings $\lambda_0, \lambda_1 \in \Lambda$ that satisfy

$$\frac{P(W=1 \mid \lambda_1)}{P(W=1 \mid \lambda_0)} > 1 \quad \text{and} \quad \frac{P(W,Z \mid \lambda_1)}{P(W,Z \mid \lambda_0)} = \frac{P(W \mid \lambda_1)}{P(W \mid \lambda_0)},$$

for each concept $Z$ that is causally separable with $W$.

**Lemma 2.4** (Unembedding–Embedding Relationship)**.** *Let $\bar{\lambda}_W$ be the embedding representation of a concept $W$, and let $\bar{\gamma}_W$ and $\bar{\gamma}_Z$ be the unembedding representations for $W$ and any concept $Z$ that is causally separable with $W$. Then,*

$$\bar{\lambda}_W^\top \bar{\gamma}_W > 0 \quad \textit{and} \quad \bar{\lambda}_W^\top \bar{\gamma}_Z = 0. \tag{2.1}$$

*Conversely, if a representation $\bar{\lambda}_W$ satisfies* (2.1)*, and if there exist concepts $\{Z_i\}_{i=1}^{d-1}$ such that each $Z_i$ is causally separable with $W$ and $\{\bar{\gamma}_W\} \cup \{\bar{\gamma}_{Z_i}\}_{i=1}^{d-1}$ is a basis of $\mathbf{R}^d$, then $\bar{\lambda}_W$ is the embedding representation for $W$.*

*Proof.* Use Theorem 2.2. The converse direction uses the fact that there is a basis consisting of representations of causally separable concepts, so (2.1) determines $\bar{\lambda}_W$ up to positive scaling. □

**Theorem 2.5** (Intervention Representation)**.** *Let $\bar{\lambda}_W$ be the embedding representation of a concept $W$. Then, for any concept $Z$ that is causally separable with $W$,*

$$P\big(Y = Y(W,1) \mid Y \in \{Y(W,0), Y(W,1)\}, \lambda + c\bar{\lambda}_W\big)$$

*is constant in $c \in \mathbf{R}$, and*

$$P\big(Y = Y(1,Z) \mid Y \in \{Y(0,Z), Y(1,Z)\}, \lambda + c\bar{\lambda}_W\big)$$

*is increasing in $c \in \mathbf{R}$.*

*Proof.* By Theorem 2.2 and Lemma 2.4. □

- The intervention notion of linear representation: adding $\bar{\lambda}_W$ changes the probability of the target concept ($W$), but not the probability of off-target concepts ($Z$).

## *3 Inner Product*

Given linear representations, we would like to measure the similarity between them, or edit concepts by projecting onto a target direction. Both require an inner product. Which one?

### Preliminaries

- The unembedding space $\Gamma$ is only an affine space, since the softmax is invariant to adding a constant. Thus, we work with $\bar{\Gamma}$, the space of differences between elements of $\Gamma$.
- Unembedding representations are only directions (unique up to positive scaling). Once we have an inner product, we define the *canonical* unembedding representation $\bar{\gamma}_W$ to be the normalized element of the cone, i.e., $\langle \bar{\gamma}_W, \bar{\gamma}_W \rangle = 1$.

Unidentifiability of the inner product. Is there a natural inner product picked out by training? No. For any invertible $A \in \mathbf{R}^{d \times d}$ and any $\beta \in \mathbf{R}^d$, consider the transformation

$$g(y) \leftarrow A\gamma(y) + \beta, \qquad l(x) \leftarrow A^{-\top}\lambda(x). \tag{3.1}$$

Then $l(x)^\top g(y) = \lambda(x)^\top \gamma(y) + \lambda(x)^\top A^{-1}\beta$. The last term does not depend on $y$, so it cancels in the softmax:

$$\frac{\exp(l(x)^\top g(y))}{\sum_{y'} \exp(l(x)^\top g(y'))} = \frac{\exp(\lambda(x)^\top \gamma(y))}{\sum_{y'} \exp(\lambda(x)^\top \gamma(y'))}.$$

- But for a fixed inner product, in general $\langle \bar{\gamma}_W, \bar{\gamma}_Z \rangle \neq \langle A\bar{\gamma}_W, A\bar{\gamma}_Z \rangle$.
- So there is no obvious reason to expect algebraic manipulations based on, e.g., the Euclidean inner product (such as cosine similarity) to be semantically meaningful.[7]

[7] Perhaps not careful enough. That a quantity is not invariant under such (gauge) transformations does not show that it is meaningless. E.g., the numerical values of physical constants may well change when we change units. Perhaps a weaker claim is true: such a quantity does not directly/literally correspond to anything significant.

### *3.1 Causal Inner Products*

Instead, we should choose the inner product more carefully: causally separable concepts should be represented as orthogonal vectors. E.g., `English`⇒`French` and `male`⇒`female` should be orthogonal.

**Definition 3.1** (Causal Inner Product)**.** A *causal inner product* $\langle \cdot, \cdot \rangle_{\mathrm{C}}$ on $\bar{\Gamma} \simeq \mathbf{R}^d$ is an inner product such that

$$\langle \bar{\gamma}_W, \bar{\gamma}_Z \rangle_{\mathrm{C}} = 0$$

for any pair of causally separable concepts $W$ and $Z$.[8]

[8] Here we assume that we already know which concepts are causally separable.

This choice turns out to unify the unembedding and embedding representations:

**Theorem 3.2** (Unification of Representations)**.** *Suppose that, for any concept $W$, there exist concepts $\{Z_i\}_{i=1}^{d-1}$ such that each $Z_i$ is causally separable with $W$ and $\{\bar{\gamma}_W\} \cup \{\bar{\gamma}_{Z_i}\}_{i=1}^{d-1}$ is a basis of $\mathbf{R}^d$. If $\langle \cdot, \cdot \rangle_{\mathrm{C}}$ is a causal inner product, then the Riesz isomorphism $\bar{\gamma} \mapsto \langle \bar{\gamma}, \cdot \rangle_{\mathrm{C}}$, for $\bar{\gamma} \in \bar{\Gamma}$, maps the unembedding representation $\bar{\gamma}_W$ of each concept $W$ to its embedding representation $\bar{\lambda}_W$:*

$$\langle \bar{\gamma}_W, \cdot \rangle_{\mathrm{C}} = \bar{\lambda}_W^\top.$$

*Proof.* Essentially by definition: by Definition 3.1, the functional $\langle \bar{\gamma}_W, \cdot \rangle_{\mathrm{C}}$ satisfies (2.1), so the converse part of Lemma 2.4 applies. □

Once we have the causal inner product, we can simply read off $\bar{\gamma}_W$ from the parameters and lift it to $\bar{\lambda}_W$, which serves as the steering vector.

### *3.2 An Explicit Form for the Causal Inner Product*

How to find the causal inner product? In principle, we could estimate the unembedding representations of many concepts, then find an inner product that makes each causally separable pair orthogonal. In practice, this is infeasible: it would take too many concepts, and each representation is hard to estimate.

Insight

> Knowing the value of a concept $W$ expressed by a randomly chosen word tells us little about the value of a causally separable concept $Z$ expressed by that word. If we learn that a randomly sampled word is French (not English), this gives us little information about whether it refers to a man or a woman.

**Assumption 3.3.** Suppose $W, Z$ are causally separable concepts and that $\gamma$ is an unembedding vector sampled uniformly from the vocabulary. Then, $\bar{\lambda}_W^\top \gamma$ and $\bar{\lambda}_Z^\top \gamma$ are independent for any embedding representations $\bar{\lambda}_W$ and $\bar{\lambda}_Z$ for $W$ and $Z$, respectively.[9]

[9] The proof of Theorem 3.4 only needs them to be uncorrelated.

This connects causal separability with something we can actually measure: the statistical dependence between words in the vocabulary.

**Theorem 3.4** (Explicit Form of Causal Inner Product)**.** *Suppose there exists a causal inner product, represented as $\langle \bar{\gamma}, \bar{\gamma}' \rangle_{\mathrm{C}} = \bar{\gamma}^\top M \bar{\gamma}'$ for some symmetric positive definite matrix $M$. If there are mutually causally separable concepts $\{W_k\}_{k=1}^{d}$ such that their canonical representations $G = [\bar{\gamma}_{W_1}, \ldots, \bar{\gamma}_{W_d}]$ form a basis for $\bar{\Gamma} \simeq \mathbf{R}^d$, then under Assumption 3.3,*

$$M^{-1} = GG^\top \quad \text{and} \quad G^\top \operatorname{Cov}(\gamma)^{-1} G = D, \tag{3.2}$$

*for some diagonal matrix $D$ with positive entries, where $\gamma$ is the unembedding vector of a word sampled uniformly at random from the vocabulary.*

*Proof.* The canonical representations are mutually orthogonal (under $\langle \cdot, \cdot \rangle_{\mathrm{C}}$) and have unit length, so $\bar{\gamma}_{W_i}^\top M \bar{\gamma}_{W_j}$ is 1 if $i = j$ and 0 if $i \neq j$. Thus, $G^\top M G = I$, i.e., $M = G^{-\top} G^{-1}$.

Next, by Theorem 3.2, $M\bar{\gamma}_{W_i}$ is an embedding representation of $W_i$. Let $\mu = \mathbb{E}[\gamma]$ be the mean of $\gamma$. By Assumption 3.3, for $i \neq j$, we have

$$\begin{aligned} 0 &= \mathrm{Cov}\big(\bar{\gamma}_{W_i}^\top M\gamma,\ \bar{\gamma}_{W_j}^\top M\gamma\big) \\ &= \mathbb{E}\Big[\big(\bar{\gamma}_{W_i}^\top M(\gamma-\mu)\big)\big(\bar{\gamma}_{W_j}^\top M(\gamma-\mu)\big)^\top\Big] \\ &= \bar{\gamma}_{W_i}^\top M\,\mathrm{Cov}(\gamma) M^\top \bar{\gamma}_{W_j} \\ &= \bar{\gamma}_{W_i}^\top M\,\mathrm{Cov}(\gamma) M \bar{\gamma}_{W_j}. \end{aligned}$$

Hence $G^\top M\,\mathrm{Cov}(\gamma) M G = D^{-1}$ for some diagonal $D$ with positive entries. Substituting $M = G^{-\top}G^{-1}$ gives $\mathrm{Cov}(\gamma) = G D^{-1} G^\top$, i.e., $G^\top \mathrm{Cov}(\gamma)^{-1} G = D$. □

Choosing the inner product

- $D$ is not unique: causal orthogonality (or uncorrelatedness) imposes $d(d-1)/2$ constraints, but a symmetric positive definite $M$ has $d(d-1)/2 + d$ degrees of freedom. So we expect $d$ remaining degrees of freedom: the diagonal entries of $D$.
- The authors simply choose $D = I_d$, which gives $M = \mathrm{Cov}(\gamma)^{-1}$.[10]
- "[A]lthough we don't have a unique inner product, we can rule out most inner products. E.g., the Euclidean inner product is not a causal inner product if $M = I_d$ does not satisfy (3.2) for any $D$." (p. 5)

[10] Namely, we set the variance of $\bar{\gamma}_{W_i}^\top M\gamma$ to be 1 for each $i$.

Unified representations. Transform the unembedding and embedding representations by $g = M^{1/2}\gamma$ and $l = M^{-1/2}\lambda$, i.e., take $A = M^{1/2}$ in (3.1).

- The embedding and unembedding representations of each concept become the same. Since $\bar{\lambda}_W = M\bar{\gamma}_W$ by Theorem 3.2,

$$\bar{l}_W = M^{-1/2}\bar{\lambda}_W = M^{-1/2} M \bar{\gamma}_W = M^{1/2}\bar{\gamma}_W = \bar{g}_W.$$

- In the transformed space, the Euclidean inner product *is* the causal inner product: $\langle \bar{\gamma}, \bar{\gamma}' \rangle_{\mathrm{C}} = \bar{\gamma}^\top M \bar{\gamma}' = \bar{g}^\top \bar{g}'$.
- With $M = \mathrm{Cov}(\gamma)^{-1}$, the coordinates of $g$ are decorrelated from each other:

$$\begin{aligned} \mathrm{Cov}(g) &= \mathbb{E}\big[M^{1/2}(\gamma-\mu)(\gamma-\mu)^\top M^{1/2}\big] \\ &= M^{1/2}\,\mathrm{Cov}(\gamma) M^{1/2} = I. \end{aligned}$$

## *4 Experiments*

- Concepts are represented as directions in the unembedding space.

- $\bar{\gamma}_W = \tilde{\gamma}_W / \sqrt{\langle \tilde{\gamma}_W, \tilde{\gamma}_W \rangle_C}$, where $\tilde{\gamma}_W$ is the mean over the $n_W$ counterfactual pairs for $W$:

$$\tilde{\gamma}_W = \frac{1}{n_W} \sum_{i=1}^{n_W} \big[\gamma(y_i(1)) - \gamma(y_i(0))\big].$$

- The estimated inner product respects causal separability.
- Concept directions act as linear probes. For $W = \texttt{French}\Rightarrow\texttt{Spanish}$, with contexts $x_j^{fr}$ and $x_j^{es}$ from French and Spanish Wikipedia, the authors expect that $\bar{\gamma}_W^\top \lambda(x_j^{fr}) < 0$, $\bar{\gamma}_W^\top \lambda(x_j^{es}) > 0$.

## *5 Questions for Discussion*

- Here, a representation of a concept $C$ is understood only as a structure that helps to measure or manipulate the probability that a token expressing $C$ appears. But one may wonder whether we can talk about representation in a wider sense: a representation of moral character should result in moral acts or moral speech, not just a higher probability that the model uses the word "morality".
- The results are only about the final layer. Can we use the same method to study the geometry of the intermediate layers? Note that for intermediate layers, we can no longer read off $\bar{\gamma}_W$ from the parameters. But we can capture more general conceptions.